
A Python-based Pipeline for Finding Motifs in Protein-Protein Interaction Networks

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1 Introduction

Networks. Biological networks are ways of representing complex biological systems using graphs of vertices and edges, with edges corresponding to relationships or interactions between objects, each object generally being represented by a vertex. Some examples are gene regulatory networks, PPI networks, and cell signaling networks. Most biological networks contain *motifs*, defined as statistically over-represented sub-graphs within the larger network graph. These motifs often reveal important features about the biological system that might not be apparent from other forms of analysis. An example of a motif commonly found in gene regulation system is the Feed-Forward Loop (FFL), which is a three-node system where two nodes (proteins) regulate a target (gene or protein) and one of the two regulators also regulates the other. Another example is the Single input module (SIM), which is a system where a single regulator controls a group of genes[1]. Commonly found motifs are depicted in Figure 1.

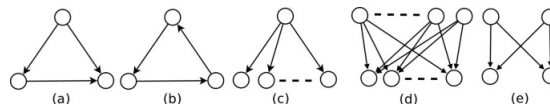


Figure 1: Representation of various motifs found in biological networks: (from left to right) Feedforward loop (FFL), 3-Cycle, Single-Input Module (SIM), Multiple-Input Module (MIM) and BiFan. Figure derived from paper by Konagurthu and Lesk[1].

Motif search. Finding motifs or sub-graphs within a network is an extension of the problem of checking if two graphs are isomorphic. Structurally, all the nodes and directed edges in one graph must have a one-to-one correspondence between the two graphs. While checking if two graphs are isomorphic is not NP-complete, determining whether a graph is isomorphic of a sub-graph of a network is. This sub-graph isomorphism problem can be solved using canonical labeling of nodes [2]. Interactions within biological networks can be studied by looking into how the motifs work in conjunction with one another[3]. Often, certain motifs are found at a higher frequency in a particular organism, revealing hints as to how the network is composed in the organism. Furthermore, the evolution of particular motifs could suggest that these patterns are more efficient in gaining a selective advantage, or how these molecular circuitries are unique in bringing together various elements in a cell for an environmental response [4]. As such, identifying motifs within a biological network is a problem of key interest in the field of computational genomics.

Algorithms for motif search. Algorithms for discovering motifs within biological networks generally can be reduced to the subgraph isomorphism check, making them NP-complete. Between this and the large size of many biological networks, motif finding is a computationally expensive problem. Despite this, many state-of-the-art algorithms exist for it. The existing algorithms are mainly differentiable in terms of three components: 1) method of generating motifs, 2) how motifs are counted, and 3) how subgraph isomorphism is checked [5]. Firstly, motifs can be generated using various pattern growth strategies, wherein a "graphlet" or "subgraph" is first constructed as a random node or edge, and is grown successively by adding more nodes and edges until a size- k motif is reached. Secondly, motifs could be counted exhaustively in the entire network, or through subgraph census. The former is more time-consuming, whereas the latter takes random samples of subgraphs from the original network using a pattern-growth strategy, and then runs motif search on the subgraphs. Finally, the subgraph isomorphism can be checked using the aforementioned canonical labelling, or by using a mapping with symmetry-breaking method. Mapping refers to ranking nodes in a network by their degree properties and then mapping these properties to those found in a motif, which allows for a fast method of checking if a node could be part of a motif. Symmetry-breaking is used to avoid checking the same sub-motif more than once within a larger motif, hence reducing the number of computations. A full summary of existing algorithms can be found in [3] and [5].

Objective. This report aims to construct a simple workflow on motif searching in a disease protein-protein interaction (PPI) network, in order to set up a standard of comparison with normal and other disease PPI networks. According to [5], a typical workflow involves the following steps: 1) finding all possible size- k subgraphs, 2) enumerating the subgraphs, and 3) determine the statistical significance of the motifs by comparison with random networks. To that end, we explore a number of strategies at each step: 1) Either predefined biological motifs, or motifs present in the network itself are used for search, 2) subgraph census is done by running multiple simulations of random sampling from the network to create a distribution of motif frequencies, and 3) motif distribution in the disease network is compared with either Barabási-Albert or Erdős-Rényi networks.

2 Methods

Dataset. The glioblastoma dataset from BioGRID [6] was used, which contains 10,356 proteins and 30,408 interactions, across 26 different experiment types and 35 different species. The vast majority of interactions are between human proteins, while a small minority are with other species. For the purpose of consistency, interactions are only considered between human proteins. Although PPI networks are considered to be undirected, we explore them as both directed or undirected graphs with separate strategies for each.

Software. All code was written in Python. The data was first read into a Pandas DataFrame to store edge lists and names of nodes. NetworkX was then used to load the data into a network structure and visualize the networks when appropriate. Random network generation, motif defining and pattern growing was also done using NetworkX. Motif searching and counting was done using grandiso's `find_motif` function, which takes a network and motif as input and returns all occurrences of that motif in the network with much shorter run-time than existing packages [7]. Finally, histograms and graphs are visualized using matplotlib.

Statistical Analysis. To generate a background frequency distribution of motifs, we used either an Erdős-Rényi graph (ER) or Barabási-Albert graph (BA) in NetworkX. ER was constructed by assuming that the degree distribution of each node in a graph is binomially distributed. On the other hand, the degree of nodes in a BA graph follow a power law, whereby nodes that already have a high degree have a higher probability of having more edges added to them. Hundred simulations of constructing ER and BA graphs, and randomly subsampling a graph of size 3000 from the glioblastoma network were performed. In each simulation, the frequency of motifs were counted and frequencies across all simulations were plotted in a histogram for comparison. The mean motif counts were compared by calculating a Z-score: $Z_{test} = \frac{\hat{f}_{PPI} - \hat{f}_{rand}}{\sigma_{rand}}$, whereby $\hat{f}_{PPI}$ is the mean motif count in the PPI network, $\hat{f}_{rand}$ is the mean motif count in the random network, and σ_{rand} is the standard deviation of motif count in the random network. Z_{test} was then used to retrieve the corresponding probability in the standard normal distribution to get a p -value.

GO Pathway Analysis. In order to assess the biological significance of motifs, the proteins found in the most common size-4 motifs were chosen for Gene Ontology (GO) and PANTHER Pathway Analysis. This was done by first condensing the glioblastoma dataset into a list of non-repeating proteins. This list was used as a "reference" set with which the motif-specific proteins were compared. For the chosen motifs, all proteins found in all occurrences in these motifs were extracted into a non-repeating list, on which enrichment analysis was done relative to all the proteins in the network using GO Biological Process [8] [9] and PANTHER Pathways [10]. Only results with FDR-corrected p -values < 0.05 were considered.

3 Implementation Details

Random Network Generation. For the directed PPI network, ER and BA graphs were generated with number of nodes set at 30. Additionally, the number of edges attached from new nodes to existing nodes, m , in the BA graph, and the probability of new edge creation, p , in the ER graph, were set to 2 and 0.102 respectively. These values were optimized for both run-time as well as matching the degree properties in the PPI network (discussed in Section 4.1).

Predefined Motif Search in Directed PPI Network. Known biological motifs were introduced in Figure 1. Among these motifs, FFL, 3-Cycle and BiFan were chosen for search as these had fixed sizes. SIM and MIM, on the other hand, could have variable numbers of inputs and outputs and were thus excluded from the current study. The chosen motifs were assigned to their own graph objects on NetworkX and fed into grandiso's `find_motif` function directly. Since this method used predefined motifs using prior knowledge, the run-time was greatly reduced.

De novo Motif Search in Undirected PPI Network. In order to generate random motifs and subgraphs with specified node size from the glioblastoma network, two types of subgraph generators were implemented: pruning and pattern-growing methods. Both implementations select a source node at random. From the source node, pruning methods generate a sub-network from a tree made from breath-first search (BFS). The sub-network is pruned until the node size matches the user designated size. As for the growing method, nodes are added one by one to the source node, searching for new nodes from the neighbors of the existing nodes. This method is adapted from [11]. Using the defined random subgraph generators, subgraphs of size 30 were sampled from the glioblastoma network and motifs were searched within the subgraph until specified percentage of edges were covered, usually between 40% to 60%, by the edges in set of discovered motifs. This motif search was simulated 50 times and the frequencies of each motif was recorded. The same ER and BA generators, and respective parameter settings used previously, were used to build random undirected networks. Grandiso's `find_motif` was used to find the motifs in the subgraphs of these random networks. The size of the subgraph of glioblastoma network and the size of the random graphs were all set to 300 nodes. Though self-interacting proteins (SIP) could have biological significance in cellular functions or signal transduction [12], self interactions or self-loop edges were not considered as NetworkX random graph generators do not create self-loops, which would have caused no motifs with self-loops to be found.

4 Results

4.1 Random networks were generated by matching graph properties with glioblastoma network.

In order to generate suitable random networks, we first assessed the properties of the glioblastoma PPI network. A randomly sampled subgraph of the network is shown in Figure 2 (first left). The degree histogram of the entire network, i.e. the frequency of edges present per node, is also shown in Figure 2 (second left). From these two images, we can see that the network appears to be relatively sparse, with few hub proteins connected to many others and a right-skewed distribution of edges such that most nodes have 1 to 3 edges connected to them. The network was found to have an average node degree of 5.87. Therefore, the parameters of the random BA and ER graphs were adjusted to achieve an average node degree of ~ 6 . Examples of such graphs are shown in Figure 2 (third and fourth). Given the sparsity of the glioblastoma network, it was reasonable to speculate that its properties would be more similar to a BA graph, which also has hub nodes that have a much higher degree than other nodes. This was further explored in the later experiments.

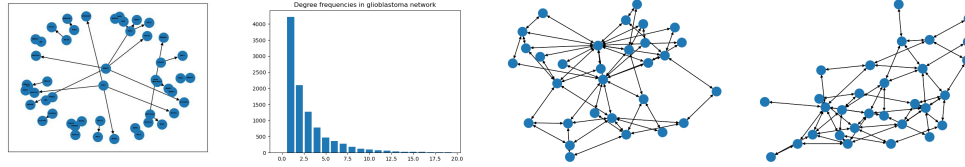


Figure 2: Properties of the glioblastoma PPI network and random BA and ER graphs. (From left to right) A subgraph sampled from the glioblastoma network, a histogram of degree frequencies of nodes found in the glioblastoma network (degree number on x-axis and frequency of number on y-axis), a random directed BA graph and a random directed ER graph.

4.2 Known biological motifs are underrepresented in glioblastoma network relative to BA graphs.

Once the parameters for generating comparable BA and ER graphs were chosen, predefined FFL, BiFan and 3-Cycle motifs were counted using the subgraph census method. Unlike an exhaustive search, which finds every single occurrence of the motif in a given network, the census method takes random subsamples from the network and counts all motif occurrences in the subsample. After running 100 simulations of generating random networks and subsamples from the PPI network, the histograms of motif frequencies were plotted (Figure 3). The results seem to suggest that all graphs showed a Gaussian-like distribution of motif frequencies, with BA graphs showing the highest counts among all the graph types.

The distributions of motif counts are computed in Table 1 and the comparison of means using Z-tests show that all motif types have lower counts in the glioblastoma network than random BA graphs. In particular, the FFL and 3-Cycle motifs are almost absent from the glioblastoma network, while BiFans are present to a moderate extent. Relative to ER graphs, however, the frequency of FFL and BiFans in glioblastoma are not significantly different, whereas 3-Cycle motifs are greatly reduced.

These observations could be due to the sparsity of the PPI network and the edge distribution in the random networks. Like BA graphs, PPI networks are thought to be scale-free, so the edges are concentrated around some hub nodes. Thus, it would be harder to find motifs with a more even edge distribution like FFLs or 3-Cycles. BiFans, on the contrary, have a more distinct hierarchy whereby some proteins are effectors and some are targets. This appears to be more easily found in directed graphs regardless of graph type. As a result, we could say that FFLs and 3-Cycles are not motifs expected in PPI networks. It might even be the case that a directed graph is not the appropriate representation for this dataset since even BiFan motifs are underrepresented relative to random graphs. After all, the source and target node were arbitrarily assigned as "Interactor A" and "Interactor B" in the dataset, which was not a definite causal association. Therefore, it was also important for us to try *de novo* motif finding in the next section by assuming an undirected graph structure.

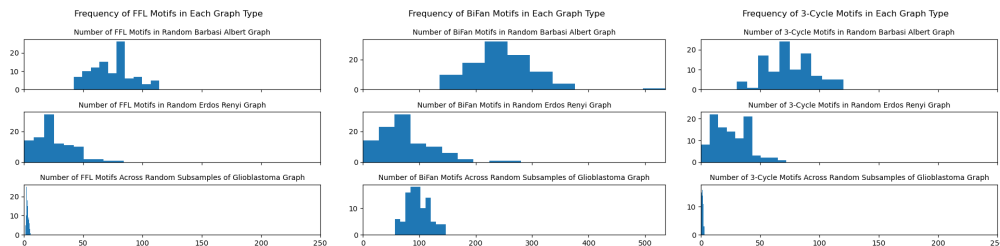


Figure 3: Motif distributions of FFLs (left), BiFans (middle) and 3-Cycles (right) from 100 simulations. Motif count is plotted on the x-axis while the frequency of the count is shown on the y-axis. The individual distributions shown are BA (top), ER (middle) and glioblastoma (bottom).

Motif	Graph Type	Mean	SD	<i>p</i> -value
FFL	BA	74.2	16.8	1.14e-5
	ER	25.1	15.4	0.07
	PPI	2.75	0.935	
BiFan	BA	243.5	57.0	0.005
	ER	74.2	47.9	0.315
	PPI	97.2	20.4	
3-Cycle	BA	73.9	19.6	1.06e-4
	ER	25.0	14.1	0.045
	PPI	1.13	0.691	
Size 3 Motif 1	BA	10388.74	970.90	1.34e-17
	ER	12276.3	550.88	2.16e-75
	PPI	2175.58	1042.57	
Size 3 Motif 2	BA	29070.3	1324.47	1.34e-17
	ER	3471.36	251.03	2.16e-75
	PPI	195.72	82.86	
Size 4 Motif 1	BA	8988349.32	19.6395621.80	3.79e-90
	ER	1397043.6	169862.90	1.7e-2
	PPI	1036867.56	1179476.06	
Size 4 Motif 2	BA	937055.96	55095.61	1.42e-64
	ER	93330.74	14512.81	4.54e-10
	PPI	4442.58	2702.94	
Size 4 Motif 3	BA	930800.0	59432.03	2.58e-55
	ER	49758.16	5022.44	1.91e-21
	PPI	2358.08	1131.84	

Table 1: Motif count distributions over 100 simulations. The result of Z-test between PPI and random network is shown under *p*-value.

4.3 Result of *de novo* motif search.

In undirected graphs, motifs with node size three can either take the shape of a line or a triangle (Figure 4). For motifs of size 4, more various shapes were generated (Figure 4). Hundred simulations of searching those motifs in random networks and glioblastoma network subgraphs, the histograms of motif frequencies were plotted as shown in Figure 4. While the histograms seem to follow normal distribution, distribution generated from glioblastoma network seem to have lower variance than the distributions generated by BA and ER. This suggests the counts of the motifs in subgraphs are occurring at a similar rate, though due to outliers, standard deviations were high. All motif types were found to have lower counts in glioblastoma network. Removing directionality in the motifs and graphs resulted in creating motifs with generic shapes, finding high motif counts. Due to this, the difference in mean motif counts in BA, ER and PPI network were significantly different (Table 1). This could be because of the fact that the glioblastoma disease network is simply a subset of all human PPIs, hence it does not have all the connections present in a normal PPI network, leading to lower counts of these motifs being found.

4.4 Size-4 motifs are part of common cancer circuitry.

Once the motifs were enumerated in the network, the most frequent size-4 motifs were chosen for analysis of biological significance (Figure 5, top row). Many proteins were found commonly among occurrences of each motif, which suggested that the motifs were repetitive in the sub-networks of specific important proteins and that the motifs could be connected to each other via shared edges. This meant that it was likely to find motifs recurring in specific biological pathways within the larger glioblastoma network and could suggest how modularity plays a role in disrupting specific molecular circuits. To that end, all the proteins that were found in every occurrence of these motifs were collected, and the enrichment of their associated Gene Ontology (GO) Biological Pathways (BP) and PANTHER Pathways was computed relative to the overall glioblastoma network.

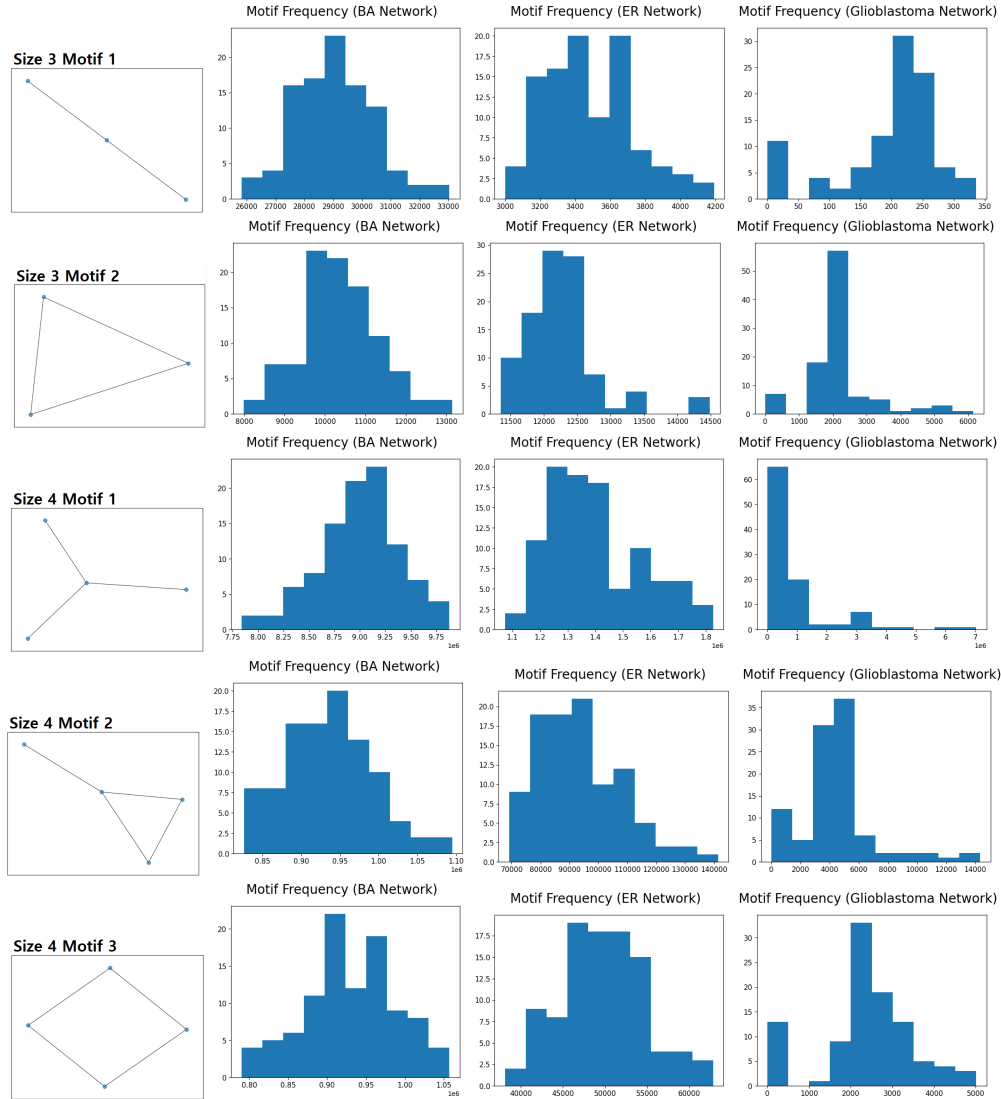


Figure 4: Size 3 and 4 motifs (column 1) frequently found in the undirected glioblastoma dataset and their frequencies in BA (column 2), ER (column 3) and glioblastoma (column 4) networks. The counts are plotted on the x-axis and their frequencies are plotted on the y-axis.

From Figure 5 (second row), we see several GO-BP terms showing high p-values with small fold enrichment, as well as several terms with high fold enrichment and lower p-values. To get the most significant and enriched terms, thresholds were defined appropriately (red dashed lines). The terms meeting surpassing these thresholds are depicted in Figure 5 (third row) along with the number of proteins found for those terms. Interestingly, we see processes relating to protein modifications and metabolism enriched for the first motif and processes relating to cell cycle and signal transduction enriched for the second motif.

A similar analysis is applied to PANTHER Pathway terms (Figure 5, fourth row). Many more terms pass the significance thresholds (Figure 5, bottom row). Interestingly, both motifs are highly enriched for CCKR signaling pathway, which is known to be found in neuroendocrine tumors as it promotes proliferation [13]. Many pathways enriched for both motifs include Wnt, PI3K, VEGF, FGF and Integrins are notable for their role in promoting epithelial-mesenchymal transition (EMT), which marks the progression of metastasis in cancers [14]. We also see some brain-specific pathways involved such as axon guidance, hypoxia response and dopamine signaling, which suggest some possible motif-specific effects in glioblastoma.

5 Conclusions

In conclusion, we have explored two methods of motif-finding in a PPI network using glioblastoma as an example. Firstly, we assumed the structure of a directed network and searched for known biological motifs, namely FFLs, 3-Cycles and BiFans. We found that these were quite common in random networks and under-represented or even absent from PPI networks. This is in line with existing literature [1] that such motifs are perhaps not unique to biological networks, although other literature supports their role in the evolution of biological circuitry [4]. We speculate that the PPI network did not support causal relationships due to the type of experimental data present, which may detect more associations between proteins rather than causal interactions. Therefore, we also used the pattern-growing and pruning methods for discovering *de novo* motifs. Although more computationally expensive, this allowed us to detect motifs that were more unique to the PPI network itself. However, these were also underrepresented relative to random networks, possibly due to the disease network being a mere subset of true human PPI networks. Some highly recurring motifs were chosen for biological analysis and there was a good correspondence of the enriched GO-BP and PANTHER pathways with existing literature on glioblastoma, as well as possible suggestions for future exploration. Thus, the motifs found could have some role in coordinating the effects of proteins in these pathways, but this requires further exploration. Along these lines, work done by Agrawal, Zitnik and Leskovec [15] suggests that smaller subgraphs form important structures that drive disease-related perturbation in a greater human PPI network, and this could generally enable link prediction to discover new proteins associated with diseases. Hence, future work can explore how glioblastoma PPI motifs map onto a normal human PPI network, in order to predict new disease-associated proteins.

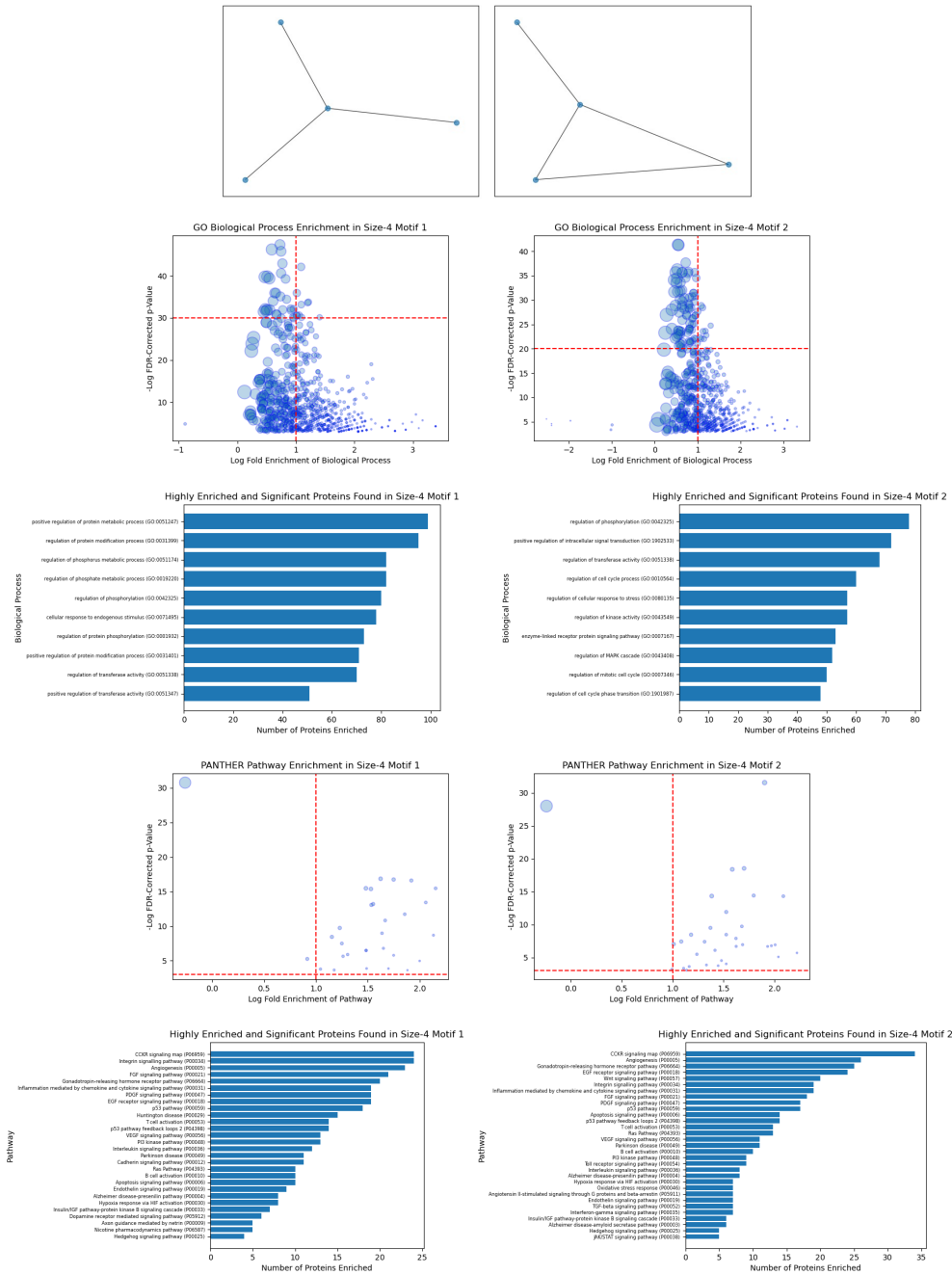


Figure 5: Size-4 motifs were chosen for further analysis using Gene Ontology (GO) and PANTHER Pathway Analysis. (Top row) Size-4 motifs with highest frequency in glioblastoma network were selected for analysis. (Second) Scatter plot of log p-value against log fold enrichment of GO Biological Process (BP) terms. Size of the dots represents the number of proteins found for these terms. Horizontal and vertical red lines represent threshold chosen to filter out most significant terms, with $p=1e-30$ on the left and $p=1e-20$ on the right. At least 10-fold enrichment of term is chosen for x-axis. (Third) Most significant GO-BP terms against the number of proteins in motifs with those terms. (Fourth) Scatter plot of log p-value against log fold enrichment of PANTHER Pathway terms. Size of the dots represents the number of proteins found for these terms. Horizontal and vertical red lines represent threshold chosen to filter out most significant terms, with $p=1e-3$ for both motifs. At least 10-fold enrichment of term is chosen for x-axis. (Bottom row) Most significant PANTHER Pathway terms against the number of proteins in motifs with those terms.

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